

Shilpa Rajesh Lingadal

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EDUCATION

Northeastern University

Master of Science in Software Engineering Systems (GPA – 3.85/4)

Boston, MA

Sep'24 - Expected Dec'26

Courses – Object Oriented Design– Java, Program Structure and Algorithms, Web Design and User Experience, Data Management and Data Design, Advanced Cloud Computing, User Experience Design and Testing

Dr. Ambedkar Institute of Technology, Visvesvaraya Technological University (VTU)

Bachelor of Engineering in Computer Science and Engineering

India

Sep'18 - Aug'22

WORK EXPERIENCE

Amazon Robotics | Software Development Engineer Co-op

Aug'25 - Dec'25

- Validated a new robotics fulfillment architecture at **158,000 units/day** with under **8% work gaps**, as measured by throughput and starvation benchmarks, by designing a multi-agent simulation engine in Java modeling drive traffic and mission types.
- Built a configurable data mining framework processing large-scale autonomous system logs to extract labeled and unlabeled performance metrics, drive behavior patterns, and congestion signals — enabling targeted sample selection across distributed compute runs.
- Decoupled **Java** simulation from **Python** analytics clients, as measured by reliable async message delivery across teams, by architecting a distributed messaging layer using **RabbitMQ**.
- Improved pipeline observability and deployment reliability by containerizing services with Docker, deploying on Kubernetes, orchestrating CI/CD pipelines, and building distributed analytics pipelines that converted simulation outputs into actionable BOM and floor layout decisions.

Infor | Software Engineer

Oct'22 - Jul'24

- Analyzed Infor Nexus – **Transport Management System(TMS) - Network Transport Visibility (NTV)** workflows, data pipelines, and system integrations to identify performance bottlenecks and improve real-time shipment visibility across supply chain operations.
- Increased enterprise client engagement by **30%**, as measured by client activity metrics across the **NTV** platform, by leading agile development of multi-modal shipment tracking features improving real-time supply chain visibility.
- Reduced API response times by **60%** (from **5s** to under **2s**), as measured by end-to-end load time benchmarks, by reengineering RESTful data retrieval logic and optimizing query execution in **Java** with **Hibernate** and **JPA**.
- Reduced system latency by **25%**, as measured by response time profiling across high-throughput data pipelines, by applying targeted backend optimizations to enterprise-scale logistics **microservices**.

Infor | Software Engineer, Associate

Apr'22 - Sep'22

- Improved database query performance and stability, as measured by reduced execution times and zero data integrity violations post-migration, by implementing query tuning, indexing strategies, and schema migrations across production systems.
- Delivered defect-free client deployments, as measured by resolution of all critical issues in QA, by conducting structured integration testing across **Java**, **Hibernate**, **JPA**, and **front-end framework layers**.
- Improved codebase maintainability and testability, as measured by a **30%** reduction in code complexity, by refactoring legacy service modules to enforce cleaner object-oriented design boundaries.

PROJECTS

SmartURL — AI-Powered URL Shortener

- Curated and preprocessed a 549,000-sample labeled dataset for ML training, built and evaluated a dual-model classification pipeline (**Random Forest 87.97%**, **Neural Network 85.47%**) using **Scikit-learn**, **Pandas**, and **NumPy**, and engineered a high-throughput serving layer with **Redis**-first caching at sub-10ms latency — demonstrating end-to-end dataset curation, model evaluation, and production ML pipeline deployment.

AI-Powered Adaptive Rate Limiter

- Achieved **200+ RPS at sub-7ms p99 latency** by engineering a distributed rate limiting service on **Linux** with **Redis**-backed shared state, three rate limiting algorithms, and a real-time anomaly detection layer using Isolation Forest — demonstrating production-grade low-latency system design.

TECHNICAL SKILLS

Languages: Java, Python, JavaScript, TypeScript, SQL, C#, HTML, CSS, C, C++, React, Angular, Flutter, Golang

Backend & Systems: Spring Boot, REST APIs, Microservices, Distributed Systems, Scalable Architecture, IPC, Agile, Scrum, SDLC, Messaging Layer (RabbitMQ, Redis), High Throughput Systems, Linux (Ubuntu/CentOS), Shell Scripting (Bash), UNIX, GraphQL, Kafka

Cloud & DevOps: AWS (Lambda, DynamoDB), Azure, Docker, Kubernetes, Container Orchestration, CI/CD Pipelines, Terraform, Packer, Unit Testing, Integration Testing, Git, pipeline orchestration, distributed compute

Databases: PostgreSQL, MySQL, MongoDB, SQL Server, Hibernate, JPA

ML & AI: TensorFlow, PyTorch, Scikit-learn, Pandas, NumPy, PyArrow, Computer Vision, Data Mining, Dataset Curation